

Lab 4, Act 1

If we knew the truth, would our trend analysis find it?

NRES 746 – name: _____ group: _____

The scenario

A state wildlife agency is planning a long-term monitoring program for a rare, secretive toad in a remote Great Basin spring complex. The management goal is to detect a population decline of 25% or more over 25 years. In Act 3 you will design and evaluate the full program. Before that, we need to know whether the statistical machinery we plan to use can be trusted at all. Real monitoring data cannot tell us, because we never know the true rate of decline. A simulation can, because we get to choose the truth.

Today's simplified world: each year a survey produces a count index y_t for years $t = 1, 2, \dots, 25$. The population declines by the same proportion every year, and each year's count differs from the trend by a random multiplicative factor. Here is the truth, which the analyst never sees:

Quantity	Value
Expected index in year 0	$N_0 = 60$
Total change from year 0 to year 25	-25% (falls to 75% of its year-0 value)
Year-to-year scatter (SD of the log of the multiplicative factor)	$\sigma = 0.2$
Surveys	one per year, $t = 1, \dots, 25$

The analyst sees only the 25 counts, fits a straight line to the *log* counts using `lm()`, and reports the slope with a 90% confidence interval. **Across many repeated monitoring programs, does that slope hit the truth on average, how variable is it, and does the interval contain the truth 90% of the time?**

Before you do anything else: close your laptop and put your phone away. Parts 1-3 and the group check-in that follows are pen-and-paper only.

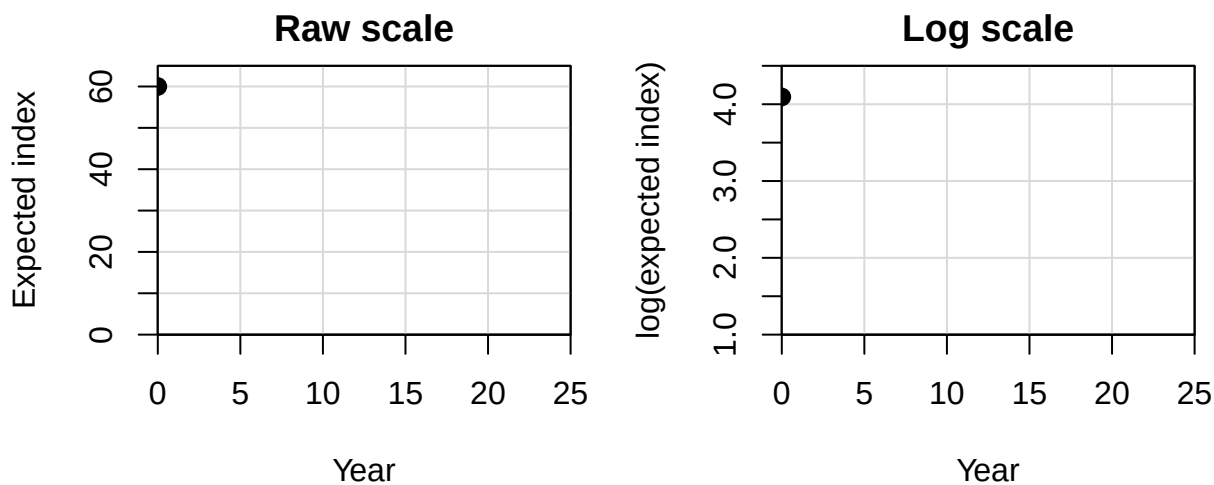
Part 1: The model (individual, ~7 min)

1a. A decline of 25% over 25 years means the index in year 25 is 0.75 times the index in year 0. If the index is multiplied by the same factor λ every year, what is λ ? Let $b = \ln \lambda$. What is b ?

$\lambda =$ _____ $b =$ _____

Show your work:

1b. Sketch the expected (median) trajectory of the index for a 25% decline and, as a dashed curve, for a 90% decline. Do this on the raw-scale axes at left, then again on the log-scale axes at right. Both start at $N_0 = 60$ in year 0 (the dot).



Which of your four curves are straight lines, and why?

1c. Write the model for the index y_t on the raw scale as N_0 times a deterministic decline, times a random factor e^{ε_t} with $\varepsilon_t \sim \text{Normal}(0, \sigma)$, independent across years. Then take logs to get a model for $\log(y_t)$. Which of the two versions is a straight-line regression model, and in what?

Part 2: What should the simulation show? (individual, ~10 min)

The analyst regresses $z_t = \log(y_t)$ on t . The least-squares slope can be written as a weighted sum of the observations,

$$\hat{b} = \sum_t w_t z_t \quad w_t = \frac{t - \bar{t}}{S_{tt}} \quad S_{tt} = \sum_t (t - \bar{t})^2,$$

where the sums run over the years surveyed. You may use two facts about the weights: $\sum_t w_t = 0$ and $\sum_t w_t t = 1$. The rules from the Week 4 WOD for expectations and variances of sums of independent random variables apply.

2a. Show that $\mathbb{E}[\hat{b}] = b$. What does this say about bias?

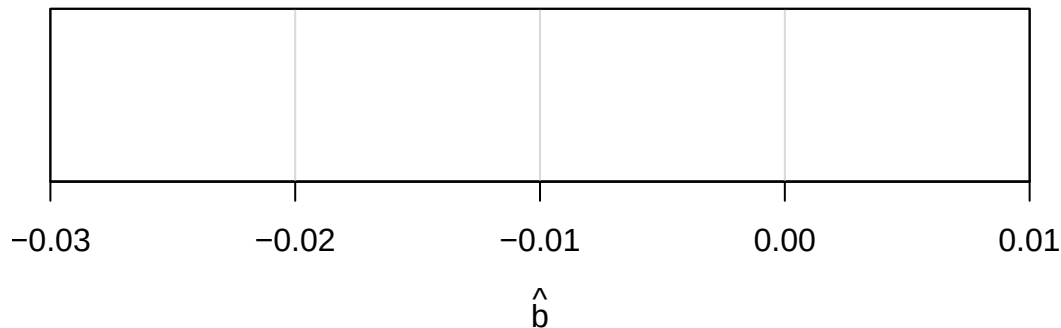
2b. Show that $\text{Var}(\hat{b}) = \sigma^2/S_{tt}$. (First show that $\sum_t w_t^2 = 1/S_{tt}$.)

2c. For annual surveys ($t = 1, \dots, 25$), $S_{tt} = 1300$. Now consider surveying only every other year ($t = 1, 3, 5, \dots, 25$, 13 surveys). Compute S_{tt} for this design, then predict the SD of $\hat{b}$ for both designs.

Annual: $S_{tt} = 1300$, $\text{SD}(\hat{b}) =$ _____

Every other year: $S_{tt} =$ _____, $\text{SD}(\hat{b}) =$ _____

2d. Suppose we simulate 2,000 monitoring programs with annual surveys and record $\hat{b}$ each time. On the axes below, mark the true value b and sketch the histogram you expect to see. Label the center and the spread on the horizontal axis.



2e. In a moment you will estimate the coverage of the 90% interval as the fraction of R simulated programs whose interval contains b . Treat each program as a Bernoulli trial with success probability $p = 0.9$. Using the rules for the variance of a mean (Week 4 WOD), what is the standard error of the estimated coverage, as a function of R ? How large must R be for that standard error to be at most 0.01?

Part 3: Plan the experiment (individual, ~3 min)

3. Write, in plain English, the step-by-step procedure for an experiment that answers the question at the end of the scenario. Do not write R code. Say what gets repeated, what you keep track of on each repeat, and what you compute at the end.

Group check-in: finalize your plan, then choose your path (group, ~7 min)

Compare your answers to Parts 2 and 3 with your group. Agree on one version of the derivation, the predicted SDs, and the plain-English procedure. Where you differ, argue it out.

Your group must now pick **one** of the two paths below for running the experiment. Both are equally valid; decide as a group before opening any laptops.

- **Path A (AI-assisted):** write a prompt describing your procedure and the model, and have an AI assistant generate the R code.
- **Path B (self-coded):** write the R code yourselves, using R's documentation and each other, but no AI tools.

Our group is using: Path A (AI-assisted) / Path B (self-coded) (*circle one*)

Laptops open now. Complete only the section below that matches your group's chosen path.

Part 4A: AI-assisted path (laptops, ~20 min)

Use these settings: $N_0 = 60$, $\sigma = 0.2$, b from 1a, and 2,000 simulated programs per design. The analysis is `lm()` of $\log(y_t)$ on t , with a 90% confidence interval for the slope from `confint()`.

Write a prompt describing your group's procedure and asking the AI to write R code that (1) runs the experiment for annual surveys ($t = 1, \dots, 25$), (2) reports the mean and SD of $\hat{b}$ and the estimated coverage of the 90% interval, (3) draws a histogram of the $\hat{b}$ values with the true b marked and the normal curve implied by your answers to 2a-2c overlaid, and (4) repeats the experiment for surveys every other year ($t = 1, 3, \dots, 25$).

Notes on the prompt you used and any changes you had to make to get the AI to do the right thing (record your prompt and what happened, not the AI's code):

Results, annual surveys. Mean of $\hat{b}$ = _____ SD of $\hat{b}$ = _____

Coverage of the 90% interval = _____ $\pm$ _____ (Monte Carlo standard error)

Results, every other year. Mean of $\hat{b}$ = _____ SD of $\hat{b}$ = _____

How do the simulated results compare with your predictions in 2a-2e and your sketch in 2d?

Reflection. Did the AI's first attempt do what your procedure said? Where did your own understanding of the model from Parts 1-2 help you catch or fix something?

Part 4B: Self-coded path (laptops, ~20 min)

Use these settings: $N_0 = 60$, $\sigma = 0.2$, b from 1a, and 2,000 simulated programs per design. The analysis is `lm()` of $\log(y_t)$ on t , with a 90% confidence interval for the slope from `confint()`.

Write the R code for your group's procedure (R's built-in help, documentation, and human-written websites and neighbors are fair game; AI tools are not). Run the experiment for annual surveys ($t = 1, \dots, 25$), then for surveys every other year ($t = 1, 3, \dots, 25$). Draw a histogram of the $\hat{b}$ values for annual surveys with the true b marked and the normal curve implied by your answers to 2a-2c overlaid.

Results, annual surveys. Mean of $\hat{b}$ = _____ SD of $\hat{b}$ = _____

Coverage of the 90% interval = _____ $\pm$ _____ (Monte Carlo standard error)

Results, every other year. Mean of $\hat{b}$ = _____ SD of $\hat{b}$ = _____

How do the simulated results compare with your predictions in 2a-2e and your sketch in 2d?

Reflection. What part of turning your procedure into working R gave your group the most trouble? How did you resolve it?

If you finish early: double σ to 0.4. Predict the SD of $\hat{b}$ and the coverage before you rerun, then check.

Group discussion and revision (~5 min)

Compare your results with a different group (ideally one that took the other path). Where the simulation disagreed with a prediction, find out why before moving on.

Feel free to revise any of your answers above based on the discussion (cross out and rewrite; I like to see your thinking evolve).

Please take a minute to record any changes you made and any new insights gained during the group discussion. You can also use this space to reflect on any areas of confusion (“muddiest points”) or topics you want to review or dust off.
